

Grid Connected Electricity Generation From Renewable Sources: Uzundere 1 63.0 MW Hydroelectric Power Plant Project, Turkey



BUREAU VERITAS CERTIFICATION

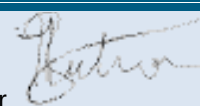
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Summary:

Bureau Veritas Certification has made the initial **and first periodic** verification of the Grid Connected Electricity Generation From Renewable Sources: Uzundere 1 63.0 MW Hydroelectric Power Plant Project, Turkey project, VCS-VCU project of Karadeniz Elektrik Uretim San. A.S. located in Uzundere Creek, Rize Province, Eastern Black Sea Region of Turkey, and applying the CDM methodology ACM0002 version 11, on the basis of UNFCCC criteria for the CDM when applicable, Verified Carbon Standard v.03, as well as criteria given to provide for consistent project operations, monitoring and reporting. UNFCCC criteria refer to Article 12 of the Kyoto Protocol, the CDM rules and modalities and the subsequent decisions by the CDM Executive Board, as well as the host country criteria.

The Uzundere-I Hydroelectric Power Plant, with a capacity of 63.068 MWm / 62.152 MWe, reduces greenhouse gas emissions that would have otherwise occurred in the absence of the project activity by avoiding electricity generation from fossil fuel sources both in the operating margin and build margin of the system. The average annual emission reductions of the proposed Project for the initial crediting period are estimated to be 87,318 tonnes of CO₂e (tCO₂e).

The verification scope is defined as a periodic independent review and ex post determination by the Validation/Verification Body of the monitored reductions in GHG emissions during defined verification period, and consisted of the following three phases: i) desk review of the project design and the baseline and monitoring plan; ii) follow-up interviews with project stakeholders; iii) resolution of outstanding issues and internal technical review followed by the issuance of the final verification report and opinion. The overall verification, from Contract Review to Verification Report & Opinion, was conducted using Bureau Veritas Certification internal procedures.

The first output of the verification process is a list of Clarification, Corrective Actions Requests, Forward Actions Requests (CL, CAR and FAR), presented in Appendix B.

In summary, Bureau Veritas Certification confirms that the project is implemented as planned and described in validated and registered project design documents. Installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately. The monitoring system is in place and the project is already generating GHG emission reductions. The GHG emission reduction is calculated without material misstatements, and the emission reductions verified totally 85,126 tons of CO₂eq for the monitoring period.

Our opinion relates to the project's GHG emissions and resulting GHG emission reductions reported and related to the valid and registered project baseline and monitoring, and its associated documents.

Reporting period	:	27/05/2010 to 1/12/2011
Baseline emissions	:	85,126 t CO ₂ equivalents.
Project emissions	:	0 t CO ₂ equivalents.
Leakage emissions	:	0 t CO ₂ equivalents.
Emission Reductions	:	85,126 t CO ₂ equivalents
		(27/05/2010 – 31/12/2010: 18,929 tCO ₂)
		(01/01/2011 – 01/12/2011: 66,197 tCO ₂)

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1 INTRODUCTION

1.1 Objective

Karadeniz Elektrik Uretim San. A.S. (KARADENİZ) has commissioned Bureau Veritas Certification to verify the emissions reductions of its VCS-VCU project Grid Connected Electricity Generation From Renewable Sources: Uzundere 1 63.0 MW Hydroelectric Power Plant Project, Turkey (hereafter called “the project”) at Uzundere creek, Rize Province, Eastern Black Sea Region of Turkey.

This report summarizes the findings of the verification of the project, performed on the basis of UNFCCC CDM Methodology ACM0002 v.11 criteria, Verified Carbon Standard (VCS) v.03, as well as criteria given to provide for consistent project operations, monitoring and reporting.

In carrying out its verification work, the Validation/Verification Body (VVB) shall ensure that the project activity complies with the requirements of paragraph 62 of the CDM modalities and procedures.

Based on the applicable requirements of paragraph 62 of the CDM modalities and procedures, this assessment shall:

- (a) Ensure that the project activity has been implemented and operated as per the registered VCS-PD and that all physical features (technology, project equipment, and monitoring and metering equipment) of the project are in place;
- (b) Ensure that the monitoring report and other supporting documents provided are complete in accordance with latest applicable version of the completeness checklist for requests for issuance of VCUs and verifiable and in accordance with applicable CDM Methodology and VCS v.03 requirements;
- (c) Ensure that actual monitoring systems and procedures comply with the monitoring systems and procedures described in the monitoring plan and the approved methodology;
- (d) Evaluate the data recorded and stored as per the monitoring methodology.

1.2 Scope and Criteria

The verification scope is defined as an independent and objective review of the project design document, the project’s baseline study and monitoring plan and other relevant documents. The information in these documents is reviewed against UNFCCC CDM Methodology ACM0002 v.11, VCS v.03 requirements, rules and associated interpretations.

The verification is not meant to provide any consulting towards the Client. However, stated requests for clarifications and/or corrective actions may provide input for improvement of the project monitoring towards reductions in the GHG emissions.

1.3 Level of assurance

The level of assurance of the Verification Report is defined as high.

1.4 Summary Description of the Project

The Grid Connected Electricity Generation From Renewable Sources: Uzundere 1 63.0 MW Hydroelectric Power Plant Project, Turkey is the implementation of a large scale hydroelectric power plant with a capacity of 63.068 MWm / 62.152 MWe on the Uzundere River, Rize, Turkey.

Project design involves the generation of electricity by utilizing the 749.41 mWC of head available at location with 2 vertical axis Pelton type turbines with 31.076 MWe capacities each. Total length of conveyance tunnel is 11,324 m. Further details regarding the technical aspects of the project design can be seen below, as given in the VCS-PD v.07, dd. 14/03/2011 of the project.

The main purpose of the project is to generate approximately 156,205 MWh/year of electricity to supply the national grid using a renewable resource and tapping the significant hydropower potential in the region. The project activity reduces greenhouse gases (GHGs) emissions that would have otherwise occurred in the absence of the project activity by avoiding electricity generation from fossil fuel sources. The average annual emission reductions of the proposed Project for the initial crediting period are estimated to be 87,318 tons of CO₂e (tCO₂e).

2 VALIDATION PROCESS, FINDINGS AND CONCLUSION

2.1 Validation Process

Please refer to the project VCS Validation Report "Validation Report for 'Grid connected electricity generation from renewable sources: Uzundere-I 63 MW Hydroelectric Power Plant Project,Turkey'", rev.02, dd. 17/03/2011 by Bureau Veritas Certification.

2.2 Validation Findings

2.2.1 Gap Validation

N/A

2.2.2 Methodology Deviations

There are no deviations from the applied project methodology, ACM0002 v.11.

2.2.3 New Project Activity Instances

Project is not a grouped project, hence N/A.

2.3 Validation Conclusion

Bureau Veritas Certification has made the validation of "Grid connected electricity generation from renewable sources: Uzundere-I 63 MW Hydroelectric Power Plant Project,Turkey". The validation has been performed on the basis of VCS criteria and host country criteria and also on the criteria given to provide for consistent project operations, monitoring and reporting.

Project participant/s used the latest tool for demonstration of the additionality. In line with this tool, the VCS-PD provides analysis of prevailing barriers to determine that the project activity itself is not the baseline scenario. By generating electricity using renewable energy, the project is seen to be likely to

result in reductions of GHG emissions partially displacing the electricity that would have been generated using fossil fuels. An analysis of the investment barrier and investment analysis demonstrates that the proposed project activity is not a likely baseline scenario. Emission reductions attributable to the project are hence additional to any that would occur in the absence of the project activity. Given that the project is implemented and maintained as designed, the project is found to be likely to achieve the estimated amount of emission reductions. The review of the Project Description (Version 01 dated 16.03.2010 until Version 07 dated 14.03.2011), the site visit and the subsequent follow-up interviews (please see the list of the persons interviewed under section 5.2 of the validation report) have provided Bureau Veritas Certification with sufficient evidence to determine the fulfillment of stated criteria. The Project Description was subsequently revised as Version 07 dated 14/03/2011 to resolve the issues that have risen during the interviews and subsequent interactions.

There are no outstanding Forward Action Requests (FARs) from the validation process.

In Bureau Veritas Certification's (BVC) opinion, the project correctly applies and meets the relevant VCS requirements and the relevant host country criteria. The validation has been based on the information made available to BVC and the engagement conditions detailed in the validation report.

3 VERIFICATION PROCESS

3.1 Method and Criteria

The verification scope is defined as a periodic independent review and ex post determination by the Validation/Verification Body of the monitored reductions in GHG emissions during defined verification period, and consisted of the following three phases: i) desk review of the project design and the baseline and monitoring plan; ii) follow-up interviews with project stakeholders; iii) resolution of outstanding issues and internal technical review followed by the issuance of the final verification report and opinion.

The overall verification, from Contract Review to Verification Report & Opinion, was conducted using Bureau Veritas Certification internal procedures.

In order to ensure transparency, a verification protocol was customized for the project, according to the version 01.2 of the Clean Development Mechanism Validation and Verification Manual, issued by the Executive Board at its 55th meeting on 30/07/2010. The protocol shows, in a transparent manner, criteria (requirements), means of verification and the results from verifying the identified criteria. The verification protocol serves the following purposes:

- It organizes, details and clarifies the requirements a VCS-VCU project is expected to meet;
- It ensures a transparent verification process where the verifier will document how a particular requirement has been verified and the result of the verification.

The completed verification protocol is enclosed in Appendix B to this report.

3.2 Document Review

The verification of the project documentation provided by the project participant is based upon both quantitative and qualitative information on emission reductions. Quantitative information comprises the reported numbers in the monitoring report submitted to the VVB. Qualitative information comprises

information on internal management controls, calculation procedures, and procedures for transfer of data, frequency of emissions reports, and review and internal audit of calculations.

In addition to the monitoring documentation provided by the project participants, the VVB reviews:

- (a) The VCS-PD (v.07), including the monitoring plan and the corresponding validation report;
- (b) Previous verification reports, if any;
- (c) Previous monitoring reports, if any;
- (d) The revised monitoring plan, if any, and its impact on the current verification;
- (e) The applied monitoring methodology;
- (d) Relevant decisions, clarifications and guidance from the VCS, CMP and the CDM Executive Board;
- (e) Any other information and references relevant to the project activity's resulting emission reductions (e.g. IPCC reports, data on electricity generation in the national grid or laboratory analysis and national regulations).

A compilation of the documents related to the verification activities have been compiled under Appendix A.

3.3 Interviews

Bureau Veritas Certification performed interviews with project stakeholders to confirm selected information and to resolve issues identified in the document review. Representatives of KARADENIZ and GAIA Finansal Danismanlik Hizmetleri Tic. Ltd. Sti. (GAIA), the consultant for the PP, were interviewed. A list of individuals that were interviewed is presented under Appendix A. The main topics of the interviews are summarized in Table 2.

Table 2 Interview topics

Interviewed organization	Interview topics
KARADENIZ	➤ Implementation and aspects of monitoring plan ➤ Technical Information About the Project ➤ Activities during the monitoring period ➤ Monitored values
Local stakeholders	➤ Local impacts of the Project ➤ Project implementation
GAIA	➤ Emission Reduction Calculations ➤ Monitoring Report ➤ All CAR's and CL's ➤ Document review issues

3.4 Site Inspections

An on-site inspection has been performed on 06/04/2012, with the participation of VVB, KARADENIZ and GAIA representatives. Interviews, inspections and document review have been performed during the site visit to verify the implementation status of the project, implementation of the monitoring methodology and the information flow for monitoring during the monitoring period. Project activities' compliance with the relevant legal regulations (health and safety practices, proper waste management, minimum water release practices, etc.) was also inspected through observations and

document review. Records and evidences of monitoring activities were reviewed with KARADENIZ and GAIA representatives to ensure authenticity and completeness of documentation/records provided.

3.5 Resolution of Any Material Discrepancy

The objective of this phase of the verification is to raise the requests for corrective actions and clarification and any other outstanding issues that needed to be clarified for Bureau Veritas Certification positive conclusion on the GHG emission reduction calculation.

Findings established during the initial and first periodic verification can either be seen as a non-fulfilment of criteria ensuring the proper implementation of a project or where a risk to deliver high quality emission reductions is identified.

Corrective Action Requests (CAR) is issued, where:

- (a) Non-conformities with the monitoring plan or methodology are found in monitoring and reporting, or if the evidence provided to prove conformity is insufficient;
- (b) Mistakes have been made in applying assumptions, data or calculations of emission reductions which will impair the estimate of emission reductions;
- (c) Issues identified in a FAR during validation or previous verifications to be verified during verification have not been resolved by the project participants.

Forward Action Requests (FAR) are issued, for actions if the monitoring and reporting require attention and/or adjustment for the next verification period.

The verification team may also use the term Clarification Request (CL), if information is insufficient or not clear enough to determine whether the applicable CDM requirements have been met.

To guarantee the transparency of the verification process, the concerns raised are documented in more detail in the verification protocol in Appendix B.

The verification report underwent an Internal Technical Review (ITR) before requesting issuance of CERs for the project activity. The ITR is an independent process performed to examine thoroughly that the process of verification has been carried out in conformance with the requirements of the verification scheme as well as internal Bureau Veritas Certification procedures.

The Lead Verifier provides a copy of the verification report to the reviewer, including any necessary verification documentation. The reviewer reviews the submitted documentation for conformance with the verification scheme. This will be a comprehensive review of all documentation generated during the verification process.

When performing an Internal Technical Review, the reviewer ensures that:

The verification activity has been performed by the team by exercising utmost diligence and complete adherence to the VCS and applicable CDM rules and requirements.

The review encompasses all aspects related to the project which includes project design, baseline, ~~additionality, monitoring plans and emission reduction calculations, internal quality assurance systems~~

of the project participant as well as the project activity, review of the stakeholder comments and responses, closure of CARs, CLs and FARs during the verification exercise, review of sample documents.

The reviewer compiles clarification questions for the Lead Verifier and Verification Team and discusses these matters with Lead Verifier.

After the agreement of the responses on the 'Clarification Request' from the Lead Verifier as well as the PP(s) the finalized verification report is accepted for further processing.

4 VERIFICATION FINDINGS

In the following sections, the conclusions of the verification are stated.

The findings from the desk review of the original monitoring documents and the findings from interviews during the follow up visit are described in the Verification Protocol in Appendix B.

The Clarification, Corrective and Forward Action Requests are stated, where applicable, in the following sections and are further documented in the Verification Protocol in Appendix A. The verification of the Project resulted in 08 Corrective Action Requests, 17 Clarification Requests, and 0 Forward Action Requests.

The CARs, CLs and FARs were closed based on adequate responses from the Project Participant(s) which meet the applicable requirements. They have been reassessed before their formal acceptance and closure.

The number between brackets at the end of each section corresponds to the CDM VVM paragraph.

4.1 Project Implementation Status

The project has been implemented in accordance with the project design document that was validated, except for the minor deviations reported below. Start date of the project activity is 27/05/2010, commissioning of the first turbine. The other unit was commissioned on 07/11/2010. Information provided in the Monitoring Report (MR) v.04 is in accordance with that stated in the validated VCS-PD v.07. Implementation status of the project structures and equipment was also verified through the Provisional Acceptance Protocols, prepared after the commissioning of each unit on aforementioned dates.

No emergencies were reported during the monitoring period. As described in the VCS-PD, calibration and maintenance responsibility for the electricity meters that are the primary source of monitoring for the fundamental parameter to be monitored belong to TEIAS. The PP is responsible to notify TEIAS in case differences in the main and backup meters are observed while TEIAS also monitors this point. No erroneous measurements have been reported; therefore no calibration or maintenance was required for the monitoring equipment.

As there was no production on July 2010, CAR04 has questioned this, and the PP stated that due to stay of execution decision by court there was no production. However, with the positive result the company restarted generating electricity.

The monitoring plan is in accordance with the approved methodology applied by the proposed VCS-VCU project activity. However certain deviations from the validated monitoring plan was observed and reported. None have any impact on the implementation of the applied methodology or on the ability of the monitoring plan to successfully monitor the emission reductions. Deviations, as reported in the MR v.04, are:

- Monthly Meter Reading Protocols that are to be signed by both parties have been determined as the source for the verification of the “*Quantity of net electricity generation supplied by the project plant/unit to the grid in year y*” parameter. However TEIAS has been migrating to remote reading arrangements and such signed protocols are now obsolete. Instead, during the monitoring period, TEIAS has been reading meters electronically on site and reporting results to the PP representatives through emails containing “*OSF Forms*”. As they represent the uncontested records of the production by the facility, PMUM records will be cross-checked with these OSF forms. No risk related to the project monitoring plan’s ability to accurately monitor and report *Quantity of net electricity generation supplied by the project plant/unit to the grid in year y* was identified.
- Monitoring method for the A_{PJ} has been defined in the approved PDD as “The readings will be done during the period when the water flow is high to have the largest reservoir area.”, however the area was calculated through the maximum water height the weir design would allow. Design maximum has been verified from approved technical project drawings.

CAR03 has been raised to ensure that the primary source for the $EG_{Facility, y}$ data and the source to cross-check this data were stated consistently in the MR. CL06 has been raised to gather information on any major overhaul activities and equipment exchange, if any. Information on the issue was added to the MR and the CL has been closed. CL08 has been raised, questioning the source of the provided information on the maximum design level stated and to justify the use of 1638 elevation as maximum. PP responded by providing approved third party project technical drawings showing that the elevation is the flood discharge level. All corrective action requests were closed as GAIA performed necessary revisions in the MR, iterating revision to 04.

The parameters required by the monitoring plan and the way the Verification Team has verified the information flow (from data generation, aggregation, to recording, calculation and reporting for these parameters including the values in the monitoring reports are described below:

(a) Parameter $EG_{Facility, y}$ – Models, specifications and serial numbers of the main and backup meters identified as the monitoring equipment for the parameter have been verified and found to be as indicated in the validated VCS-PD and VCS-MR. PMUM (Market Financial Settlement Center) records (main) and OSF forms (crosscheck) have been the source of monitoring for this parameter. The PP can access PMUM data online through a login and password provided exclusively to them by PMUM. PMUM screens clearly depict all energy generation and consumption values, on a daily basis as well as monthly totals. Net generation values were reached by subtracting the consumption of the plant from the gross amount of electricity supplied to the grid. Values reported on these screens have been cross-checked with the monthly reading records of TEIAS, submitted to KARADENIZ in OSF Forms, through e-mails from the district representatives of TEIAS. Original mails have been seen by the verification team and OSF forms were verified on site. Samples of the mails, all OSF Forms and PMUM screens have also been kept for record keeping. As indicated earlier no maintenance, calibration activities by TEIAS was performed, during the monitoring period. Based on the records, GAIA has prepared a spreadsheet to report the generation values and performed the emission reduction calculations. Reported amount of total net electricity generated in the monitoring period is indicated as 33,826.52 MWh for 2010 and 118,294.09 MWh for 2011, a total of 152,120.61 MWh.

(b) Parameter Cap_{PJ} – Total installed capacity of the project is 62.152 MWe, as indicated by the project license. Equipment and their capacities have been verified on site through the equipment placards and were cross-checked with equipment specifications as well as the provisional acceptance protocols for the plant. No changes were made to the equipment or their capacities.

(c) Parameter A_{PJ} – Area of the regulation reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full has been determined through the topographical drawings and 3D models, assuming a water height equivalent to the theoretical design maximum of the weir i.e. 1638m elevation. Actual water level was lower than this value hence the resulting area value is fairly conservative. Visual inspections on the reservoir area as well as interviews indicated no change in the physicality of the weir or reservoir boundaries. Reported value for the parameter is 7,275.54 m².

Project is compliant with the laws and regulations applicable to it. Minimum amount of water released to the river bed at all times to protect the local flora and fauna is monitored through an AGI (Flow Observation Station) that is controlled and monitored by DSI (State Hydraulic Works). KARADENİZ's compliance was verified through sampled AGI reports. KARADENİZ complies with other applicable regulations on topics such as water quality, air quality and noise pollution (monitored through measurements performed by third party), waste management (receipts of lawful disposal from authorized waste collectors), social security of workers (SGK records), health and safety implementations (EKAT certificates and availability of protection equipment) as verified by sampling through relevant records.

4.2 Accuracy of GHG Emission Reduction or Removal Calculations

In line with the applied methodology the baseline emission are calculated according to:

$$BE_y = EG_{PJ,y} \times EF_{grid, CM, y}$$

Where:

BE_y Baseline emissions in year y (tCO₂)

$EG_{PJ,y}$ Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in year y(MWh/yr)

$EF_{grid, CM, y}$ Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “*Tool to calculate the emission factor for an electricity system*”

y Refers to a given year

Emission reductions for the project have been calculated in line with the requirements of the applied methodology ACM0002 v.11:

$$\begin{aligned} BE_{2010} &= EG_{PJ,2010} \times EF_{grid, CM, 2010} \\ &= 33,826.52 \text{MWh} \times 0.5596 \text{ tCO}_2/\text{MWh} \\ &= \mathbf{18,929.32 \text{ tCO}_2} \end{aligned}$$

$$\begin{aligned} BE_{2011} &= EG_{PJ,2011} \times EF_{grid, CM, 2011} \\ &= 118,294.09 \text{MWh} \times 0.5596 \text{ tCO}_2/\text{MWh} \\ &= \mathbf{66,197.37 \text{ tCO}_2} \end{aligned}$$

$$\begin{aligned} BE_{2010} + BE_{2011} &= \mathbf{18,929.32 + 66,197.37} \\ &= \mathbf{85,126.69 \text{ tCO}_2} \end{aligned}$$

In line with the applied methodology the project emission are assumed to be 0 since the power density of the project is greater than 10 W/m² as the methodology dictates. Power density has been calculated according to:

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$$

Where:

PD = Power density of the project activity, in W/m²
 Cap_{PJ} = Installed capacity of the hydro power plant after the implementation of the project activity (W)
 Cap_{BL} = Installed capacity of the hydro power plant before the implementation of the project activity (W). For new hydro power plants, this value is zero
 A_{PJ} = Area of the reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (m²)
 A_{BL} = Area of the reservoir measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m²). For new reservoirs, this value is zero

Cap_{PJ} = 62,152,000 W
 Cap_{BL} = 0 (Justification: The project is a new hydro power plant)
 A_{PJ} = 7,275.54 m² (Reservoir area for 1638m elevation)
 A_{BL} = 0 (Justification: The project is a new hydro power plant)

Therefore;

PD = (62,152,000 – 0) / (7,275.54 – 0) = 8,542 W/m² > 10, hence:

PE_{2010/2011} = 0 tCO₂/year

In line with the guidance of the applied methodology no leakage emissions were considered,

LE_{2010/2011} = 0 tCO₂

Also, according to ACM0002/Version 11 the emission reductions in year y should be calculated as the following formula

$$ER_y = BE_y - PE_y - LE_y$$

Where:

ER_y = Emission reductions in year y (t CO₂e/yr)
 BE_y = Baseline emissions in year y (t CO₂e /yr)
 PE_y = Project emissions in year y (t CO₂/yr)
 LE_y = Leakage emissions in year y (t CO₂/yr)

Since PE_y and LE_y are assumed to be 0, emission reductions are equal to baseline emissions.

$$ER_{2010} = BE_{2010}$$

$$ER_{2011} = BE_{2011}$$

$$ER_{2010} = 18,929.32 \text{ tCO}_2$$

$$ER_{2011} = 66,197.37 \text{ tCO}_2$$

(Emission reductions generated during the initial monitoring period 27/05/2010 – 01/12/2011)

The data monitored during the monitoring period is summarized in below:

Table 3 Electricity Generated and Emission Reductions achieved by project.

Uzundere-I HEPP Emission Reductions – 2010				
Date	Net Electricity Generation (kWh)	Electricity Traction from Grid (kWh)	Net Electricity Supplied to the Grid (kWh)	Emission Reductions (tCO ₂)
Jan	0	0	0	0,00
Feb	0	0	0	0,00
Mar	0	0	0	0,00
Apr	0	0	0	0,00
May	1.807,51	73,756	1.733,75	970,21
Jun	12.738,40	0,586	12.737,81	7.128,08
Jul	0	0	0	0,00
Aug	841,941	2,965	838,976	469,49
Sep	4.116,73	17,668	4.099,06	2.293,83
Oct	5.478,10	28,499	5.449,61	3.049,60
Nov	4.959,56	33,78	4.925,78	2.756,47
Dec	4.079,22	37,69	4.041,53	2.261,64
Total	34.021,46	194,94	33.826,52	18.929,32

Uzundere-I HEPP Emission Reductions – 2011				
Date	Net Electricity Generation (kWh)	Electricity Traction from Grid (kWh)	Net Electricity Supplied to the Grid (kWh)	Emission Reductions (tCO ₂)
Jan	3.155,15	45,77	3.109,38	1.740,01
Feb	2.643,03	43,39	2.599,64	1.454,76
Mar	3.680,14	43,61	3.636,53	2.035,00
Apr	11.341,81	33,64	11.308,17	6.328,05
May	27.078,12	14,4	27.063,72	15.144,86
Jun	23.851,05	10,94	23.840,11	13.340,93
Jul	10.612,55	23,12	10.589,43	5.925,85
Aug	7.539,66	25,07	7.514,59	4.205,16
Sep	6.258,94	23,89	6.235,05	3.489,13
Oct	12.271,41	22,01	12.249,40	6.854,76
Nov	5.610,88	36,43	5.574,45	3.119,46
Dec	4.615,49	41,87	4.573,62	2.559,40
Total	118.658,23	364,14	118.294,09	66.197,37

4.3 Quality of Evidence to Determine GHG Emission Reductions or Removals

A complete set of data for the specified monitoring period for all of the monitored parameters was available for verification.

Regarding ex-post monitored parameters:

- $EG_{\text{facility},y}$: Monitored by approved, sealed equipment that are in line with the validated VCS-PD (ACTARIS SL7000 series with S/Ns: 53064430 – 53064429). All QA/QC procedures relevant have been adhered to and no events related to the equipment were identified or reported. No discrepancies between the readings of the two meters were observed by the verification team, during the site-visit. The meters were purchased for the project and the supplier company (ACTARIS) commits that these meters fully conforms or exceeds all relevant IEC standards including those dealing with electronic metering equipment IEC61036 for class 1 equipment and IEC60687 for class 0.2S accuracy equipment. There is no possibility for human error in the measurement of the electricity as it was performed by dedicated TEIAS technicians using appropriate equipment. All the measurements and calculations are done via tested meters under TEIAS responsibility. Meter readings are continuously monitored by TEIAS and plant personnel for any discrepancies. All records produced (OSF Forms, PMUM screens) are official and final with regards to the values identified. Calculations based on values acquired from and cross-checked by these sources were checked by the verification team and found to be correct and conservative.
- Cap_{PJ} : Verified on-site, through the equipment placards and cross-checked with the project EMRA generation license and Provisional Acceptance Protocols (dd. 27/05/2010 and 07/11/2010) against manual errors. No discrepancies were observed.
- A_{PJ} : Verified through the topographical drawings of the project and cross-checked with a 3D model of the reservoir area. Resulting value is based on the theoretical maximum water level that the weir is designed to hold hence it is highly conservative compared to the actual situation.

Only other relevant parameter for the emission reduction calculations is the ex-ante fixed grid combined emission factor determined in the validated VCS-PD and valid for the duration of the entire crediting period. Value indicated in the VCS-MR v.04 and used in the calculations by GAIA is 0.5596 tCO₂/MWh and is in line with the validated value from the VCS-PD v.07.

4.4 Management and Operational System

Management system and monitoring responsibilities are in line with the implementation described in the validated VCS-PD and the VCS-MR v.04 and it is the verification team's opinion that it is suitable as well as capable of successfully monitoring the parameters set forth by the monitoring plan and emission reductions.

5 VERIFICATION CONCLUSION

Bureau Veritas Certification has performed the initial and first periodic verification of the Uzundere-I Hydroelectric Power Plant Project in Turkey, which applies the methodology ACM0002 version 11. The verification was performed based on the requirements set by the VCS v.03, CDM Methodology and relevant guidance provided by CMP and the CDM Executive Board.

The verification consisted of the following three phases: i) desk review of the project design and the baseline and monitoring plan; ii) follow-up interviews with project stakeholders; iii) resolution of outstanding issues and the internal technical review followed by the issuance of the final verification report and opinion.

The management of Karadeniz Elektrik Uretim San. A.S. is responsible for the preparation of the GHG emissions data and the reported GHG emissions reductions of the project on the basis set out within the project Monitoring Plan indicated in the validated VCS-PD. The development and maintenance of records and reporting procedures in accordance with that plan, including the calculation and determination of GHG emission reductions from the project, is the responsibility of the management of the project.

Bureau Veritas Certification verified the VCS Monitoring Report version 04 for the reporting period as indicated below. Bureau Veritas Certification confirms that the project is implemented as described in validated project design documents. Installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately. The monitoring system is in place and the project is generating GHG emission reductions.

Bureau Veritas Certification can confirm that the GHG emission reduction is calculated without material misstatements. Our opinion relates to the project's GHG emissions and resulting GHG emissions reductions reported and related to the valid and validated project baseline and monitoring, and its associated documents. Based on the evidence and information that are considered necessary to guarantee that GHG emission reductions are appropriately calculated, Bureau Veritas Certification confirms the following statement:

Reporting period: From 27/05/2010 to 01/12/2011

GHG Emission Reductions or Removals	tCO ₂ e
Baseline Emissions	85,126
Project Emissions	0
Leakage	0
Net GHG emission reductions or removals	85,126

APPENDIX A:

A.1. Documents provided by KARADENIZ & GAIA that relates directly to the GHG components of the project and other reference documents are given below:

- a) VCS PD ver.07, dd. 14/03/2011;
- b) Validation Report for “Grid connected electricity generation from renewable sources: Uzundere-I 63.0 MW Hydroelectric Power Plant Project, Turkey”, Rev.02, dd. 17/03/2011;
- c) VCS MR ver.01, dd. 29/02/2012;
- d) VCS MR ver.02, dd. 15/06/2012;
- e) VCS MR ver.03, dd. 16/08/2012;
- f) Uzundere-I Verification (27/05/2010-01/12/2011) ER Calculation Excel Workbook, v01;
- g) Uzundere-I Verification (27/05/2010-01/12/2011) ER Calculation Excel Workbook, v02;
- h) PMUM screens for 27/05/2010-01/12/2011;
- i) OSF Forms for 27/05/2010-01/12/2011;
- j) Ministry of Energy and Natural Resources Provisional Acceptance Protocol, dd. 27/05/2010;
- k) Ministry of Energy and Natural Resources Provisional Acceptance Protocol, dd. 07/11/2010;
- l) Energy Market Regulatory Authority Generation License, # EU/902-4/710, dd. 31/08/2006;
- m) Reservoir area topographical drawing and dwg model;
- n) # 53-399 AGI Station data, # 61-339 AGI station data;
- o) Dokay Muhendislik Noise Analysis, # DKL-11/G246, dd. 19/02/2011; #DKL-11/G269, dd. 23/05/2011; #DKL-11/G297, dd. 27/08/2011; # DKL-11/G325, dd. 16/11/2011.
Dokay Muhendislik Water Analysis, #DKL-11/S2733, dd. 26/05/2011; # DKL-11/S2921, dd. 28/09/2011;
Dokay Muhendislik Air Analysis, # DKL-11/H564, dd. 23/05/2011; # DKL-11/H591, dd. 27/08/2011
- p) CDM Validation Verification Manual v.01.2;
- q) CDM Methodology ACM0002, v12;
- r) Verified Carbon Standard v.03 Program Guide and Standard.

A.2. List persons interviewed during the verification and site visit, or persons that contributed with other information that are not included in the documents listed above.

- a) Serkan DUNDAR – GAIA Carbon Finance;
- b) Taner SENGONUL – SANKO HEPP Investment Assistant Coordinator;
- c) Ismail Hakki ABALI – Plant Manager,
- d) Zafer OZER – Powerhouse operator, local;
- e) Kamil CIHATSARI – Plant Engineer;
- f) Ibrahim DURNA – Plant Engineer, local.

A.3. Verification Team:**Internal Technical Reviewer: Burcu Mutman**

Bureau Veritas Certification Internal Technical Reviewer

Lead Verifier: Mr. Mehmet Kumru – Environmental Engineer

Bureau Veritas Certification SAS – Lead Verifier

Verifier: Mr. Mustafa Unal – Metallurgical and Materials Engineer

Bureau Veritas Certification - Verifier

Mustafa Unal is a Metallurgical and Materials Engineer and an auditor for environment, safety and quality management systems. He has worked in automotive and civil aviation industries. He has participated online seminars in the Gold Standard Academy in 2011 and is a CDM Verifier.

APPENDIX B:

VERIFICATION PROTOCOL (revision 06)

Table 1 Verification requirements based on the Clean Development Mechanism Validation and Verification Manual (Version 01.2)

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
1 Compliance of the monitoring report with the guidelines for completing the monitoring report form					
a Brief description of the project activity					
a. Is the description of the project activity to be presented in this section a brief summary of the detailed description given in the section .B.1 Implementation status of the project activity?	EB 54	Ann 34	A summary description of the project is available in section 1.1 of the VCS Monitoring Report (MR) v.01	OK	OK
b. Does this description include:	EB 54	Ann 34			
i. Purpose of the project activity and the measures taken to reduce greenhouse gas emissions?	EB 54	Ann 34	Please indicate how the project reduces greenhouse gasses.	CL01	OK
ii. Brief description of the installed technology and equipments;	EB 54	Ann 34	Please provide information on implemented technology and installed hardware.	CL02	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
iii. Relevant dates for the project activity (e.g. construction, commissioning, continued operation periods, etc.)?	EB 54	Ann 34	Please briefly indicate dates of major events in the project's implementation.	CL03	OK
iv. Total emission reductions achieved in this monitoring period?	EB 54	Ann 34	Kindly include information on the amount of emission reductions achieved during this monitoring period."	CL04	OK
b Project participants					
a. Are the project participants listed?	EB 54	Ann 34	Project participants have been tabulated under section 1.3 as: ▪ <i>Karadeniz Elektrik Uretim A.S.(project proponent);</i> ▪ <i>GAIA Carbon Finance (carbon consultant)</i> As indicated by the reporter, GAIA is not a project proponent. Kindly move GAIA under section 1.4 of the monitoring report v.01	CL05	OK
c Location of project activity					
a. Is complete information of the location of the project activity: town, city, country and GPS coordinates.provided?	EB	Ann 34	Project is located in the Uzundere creek, Rize province of Turkey.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
	54		Powerhouse coordinates were given, in line with the project PD v.07 as 40°50'16" - 40° 54' 04" N and 40° 45' 4" - 40° 49'42" E		

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
d Technical description of the project					OK
a. Are a description of the technology applied in the project activity and detailed technical process, including diagrams provided?	EB 54	Ann 34	Please see CL02	CL02	OK
e Title, reference and version of the baseline and monitoring methodology applied to the project activity					
a Are the complete reference of the methodology applied and tools whenever is applicable included?	EB 54	Ann 34	Applied methodology is given as ACM0002 v.12.2.0, which is inconsistent with the project PD v07. Please clarify.	CAR01	OK
			"Tool to calculate the emission factor for an electricity system" v.02 is applied	OK	OK
f Registration date of the project activity					
a Is the registration date of the project activity provided?	EB 54	Ann 34	Please see CL03	CL03	OK
g Crediting period of the project activity and related information (start date and choice of crediting period)					
a Does the description also include changes to the start date of the crediting period post-registration that have been accepted by the Board, when	EB	Ann 34	N/A	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
applicable?	54				
h Name of responsible person(s)/entity(ies)					
a Is the contact information of the person(s)/entity(ies) responsible for completing the monitoring report form (CDM-MR) provided?	EB 54	Ann 34	The MR was prepared by Mr. Gurkan BAYRAKTAR from GAIA, as indicated on the cover.	OK	OK
i Implementation status of the project activity					
a Does this section include a description of the implementation and operational status of the project as of this monitoring period in accordance with the latest version of the CDM Validation and Verification Manual (CDM-VVM)?	EB 54	Ann 34	A description of the implementation status of the project has been provided. Please see below questions for further detail.	OK	OK
b Does the description include inter alia:	EB 54	Ann 34			
i The starting date of operation of the project activity? For project activities that consist of more than one site, the report shall clearly describe the status of implementation and starting date of operation for each site. For CDM project activities with phased implementation, the report shall indicate the progress of the proposed CDM project activity achieved in each phase.	EB 54	Ann 34	Project start date has been indicated as 27/05/2010, commissioning of the first turbine of two. This was verified through the Temporary Acceptance Protocol for that unit.	OK	OK
ii The information regarding the actual operation of the project activity during this monitoring period.	EB	Ann	Please provide information on any major overhaul	CL06	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
including information on special events, for example overhaul times, downtimes of equipment, exchange of equipment, etc?	54	34	activities and kindly indicate any critical equipment exchange that was performed during this monitoring period.		
iii A brief description of: (i) events or situations that occurred during the monitoring period, which may impact the applicability of the methodology, and (ii) how the issues resulting from these events or situations are being addressed?	EB 54	Ann 34	It has been indicated that in this monitoring period no emergencies or any abnormal situations were observed.	OK	OK
j Revision of the monitoring plan					
a Is it indicated if the monitoring plan has been revised?.	EB 54	Ann 34	No deviations from the monitoring plan were experienced.	OK	OK
b Is the date of approval, if revised, included?	EB 54	Ann 34	N/A	OK	OK
k Request for deviation applied to this monitoring period					
a Is any deviation applied to this monitoring period indicated?	EB 54	Ann 34	No deviations from the monitoring plan were indicated.	OK	OK
b Is the reference number, if any deviation applied, included?	EB	Ann 34	N/A	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
	54				

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
I Notification or request of approval of changes					
a Is any notification or request of approval of changes from the project activity as described in the registered CDM-PDD indicated?	EB 54	Ann 34	No notification or request of approval of changes was indicated.	OK	OK
b Is the date of approval, if applicable, included?	EB 54	Ann 34	N/A	OK	OK
m Description of the monitoring system					
a Is a description of the monitoring system provided?	EB 54	Ann 34	A description of the monitoring plan was provided in section 3.3 of MR v.01	OK	OK
b Does his section include data collection procedures (information flow including data generation, aggregation, recording, calculation and reporting), organizational structure, roles and responsibilities of personnel, and emergency procedures for the monitoring system?	EB 54	Ann 34	The description includes procedures on how the data for electricity generated by the plant will be gathered and the relevant roles and responsibilities.	OK	OK
			Section includes procedures, etc. only for electricity generated parameter. Please also include procedures, roles and responsibilities for the other parameters that are monitored	CAR02	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			Reference to ACM0002 in page 9 of MR v01 contains an version value that is inconsistent with the applied version of the methodology, please clarify. Please ensure that this inconsistency is revised throughout the Monitoring Report	CL10	OK
c Does this include line diagrams showing all relevant monitoring points?	EB 54	Ann 34	N/A	OK	OK
n Data and parameters					
a Does this section include parameters used to calculate baseline, project, and leakage emissions as well as other relevant parameters required by the approved methodology and the monitoring plan; and specific information on how data and parameters have been monitored during the monitoring period?	EB 54	Ann 34	In accordance with the applicable methodology, ACM0002 v.11 and the related tools: ▪ $EG_{Facility, y}$; ▪ Cap_{PJ}; ▪ A_{PJ}; Are the paremeters to be monitored for emission calculations, and have been included in MR v01 section 3.2.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			Kindly clarify and ensure consistency of methodology for monitoring $EG_{Facility, y}$ parameter, since various references in the MR v.01, define Monthly Meter Reading Protocols and PMUM records as main source of data and cross-check source, in a varying manner. Please mind that consistency with the registered project PD v.07 must also be ensured.	CAR03	OK
b Are data that is determined only once for the crediting period but are used after registration of the project activity included here under section D.1.?	EB 54	Ann 34	The combined emission factor of the project grid system that has been calculated ex-ante in project PD v.07 as 0.559 has been included in section 3.1, and will be valid for the duration of the crediting period.	OK	OK
c For each parameter the following information, using the tables provided, is provided:					
i Value of monitored parameter in the period for the purpose of calculating emission reductions? To report multiple values, a table may be used and included in this monitoring report or include references to spreadsheet. For default value (such as an IPCC value), where it is ex-post confirmed, the most recent value shall be applied.	EB 54	Ann 34	The values of monitored parameters have been given in section 3.2 of MR v01 as: ▪ $EG_{Facility, 2010}$: 34,021.46 MWh; $EG_{Facility, 2011}$: 118,658.23 MWh; ▪ Cap_{PJ}: 63 MW; ▪ A_{PJ}: 7,275.54 m2	OK	OK
			Kindly report value of monitored parameter for Cap_{PJ} and A_{PJ} in line with the provided frequency,	CL09	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			i.e. yearly		

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
ii Description of the equipment used to monitor each parameter, including details on accuracy class, and calibration information (frequency, date of calibration and validity), if applicable as per monitoring plan?	EB 54	Ann 34	The only parameter to be monitored through equipment is the $EG_{Facility, y}$ value. This value is monitored through two meters (one main one control, both ACTARIS SL761B071 model) with S/Ns: 53064430 and 53064429 respectively. Devices are of 0.2S accuracy class.	OK	OK
			Kindly revise parameter $EG_{Facility, y}$ to include information on calibration procedures for the monitoring devices.	CL07	OK
iii Measuring and recording method: how the parameters are measured/calculated, specifying the measurement and recording frequency?	EB 54	Ann 34	Measurement methods and relevant frequencies have been provided for all three of the parameters monitored.	OK	OK
			Please provide a source for the data provided for the A_{PJ} parameter's monitored value to verify its validity. Kindly justify the use of 1638 elevation.	CL08	OK
iv Source of data: logbooks, daily records, surveys, etc?	EB 54	Ann 34	Please see CAR03 and CL08	CAR03 CL08	OK
v Where relevant, the calculation method of the parameter?	EB	Ann 34	The calculation method for $EG_{Facility, y}$ has been given and N/A for the other two parameters.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
	54				

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
vi The QA/QC procedures applied (if applicable per monitoring plan)?	EB 54	Ann 34	QA/QC procedures were noted where applicable.	OK	OK
vii Include information about appropriate emission factors, IPCC default values and any other reference values that have been used in the calculation of emission reductions?	EB 54	Ann 34	Only emission factor or default value to be utilized is the combined margin emission factor for the Turkish grid which was calculated ex-ante in registered project PD v07	OK	OK
o Baseline emissions calculation					
a Does this section include all formulae used and description to calculate the baseline emissions applying actual values?	EB 54	Ann 34	Relevant formulae have been provided and applied with actual values for the monitoring period.	OK	OK
			Please provide Monthly Meter Reading Protocols for verification.	CAR04	OK
			Parameter defined has a yearly value, therefore please provide calculations, if applicable, and results for each year in the monitoring period separately for verification consideration.	CL11	OK
			Please review sentence in parenthesis “Since baseline emission calculations were realized in the ex-ante manner...” on page 14 of MR v01	CL13	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
b Was a table used and included in this monitoring report or include references to spreadsheet?	EB 54	Ann 34	A table was provided in section 4.1 of MR v01. However, table title and the footnote of page 14 have contradicting information with regards to the source of the data. Please clarify.	CL12	OK
			Please provide an excel worksheet clearly depicting electricity generation values with the relevant source, any formulae applied, emission reduction calculations etc.	CAR05	OK
p Project emissions calculation					
a Does this section include all formulae used and description to calculate the project emissions applying actual values?	EB 54	Ann 34	Relevant formulae are included in section 4.2 of MR v01. Please see CL11	CL11	OK
b Was a table used and included in this monitoring report or include references to spreadsheet?	EB 54	Ann 34	N/A	OK	OK
q Leakage calculation					
a Does this section include all formulae used and description to calculate the leakage applying actual values?	EB 54	Ann 34	Leakage is considered 0 in line with the applicable methodology.	CL11	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			Please see CL11		

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
b Was a table used and included in this monitoring report or include references to spreadsheet?	EB 54	Ann 34	N/A	OK	OK
r Emission reductions calculation/table					
a Does this section include the formulae used to calculate the emission reductions and the total of the emission reductions achieved during the monitoring period?	EB 54	Ann 34	Formulae used for the calculation of emission reductions have been included Please see CAR05	CAR05	OK
i Total baseline emissions:	EB 54	Ann 34	Please see CAR05	CAR05	OK
ii Total project emissions:	EB 54	Ann 34	Please see CAR05	CAR05	OK
iii Total leakage:	EB 54	Ann 34	Please see CAR05	CAR05	OK
iv Total emission reductions:	EB 54	Ann 34	Please see CAR05	CAR05	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
s Comparison of actual emission reductions with estimates in the CDM-PDD					
a Does this section include a comparison of actual values of the emission reductions achieved during the monitoring period with the estimations in the registered CDM-PDD?	EB 54	Ann 34	Project is a VER project under VCS and this is not a requirement. N/A	OK	OK
t Remarks on difference from estimated value in the PD					
a Is an explanation of the cause of any increase in the actual emission reductions achieved during the current monitoring period (e.g. higher water availability, higher load plant factor, etc), including all information (i.e. data and/or parameters) that is different from that stated in the registered CDM-PDD provided?	EB 54	Ann 34	Realized reductions are less then estimated for the same period of monitoring, hence: N/A	OK	OK
2 Project implementation in accordance with the registered project design document					
a Are all physical features of the proposed CDM project activity proposed in the registered PDD in place?	VVM	196	It was verified through the site visit that the project physical features are in line with the registered PD	OK	OK
b Have the project participants operated the proposed CDM project activity as per the registered PDD?	VVM	196	PP has operated the project per registered PD	OK	OK
c Was an on-site visit conducted?	VVM	196	Yes, a site visit was conducted on 06/04/2012	OK	OK
d If not, justify the rationale of the decision.	VVM	196			OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
e Does the implementation or operation of CDM project activity conform with the description contained in the registered PDD?	VVM	197	PP has operated the project per registered PD	OK	OK
f If not, which are the potential impacts due to these changes, according to the relevant guidelines established by the Executive Board (EB48-§73)?	VVM	197	N/A	OK	OK
g Was any change identified close to the boundary of the project activity but outside it?	VVM	197	No changes were observed	OK	OK
h If yes, which are the potential impacts due to these changes?	VVM	197	N/A	OK	OK
i Was a notification or a request for approval of changes from the project activity as described in the registered PDD submitted prior to the conclusion of the verification/certification for the corresponding?	VVM	197	No such request was issued.	OK	OK
3 Compliance of the monitoring plan with the monitoring methodology					
a Is the validated monitoring plan in accordance with the approved methodology applied by the proposed CDM project activity?	VVM	200	Monitoring plan is in line with the requirements of the applied methodology ACM0002 v.11	OK	OK
b If no, was a request for revision of the monitoring plan was done? (The DOE may request for revision of the monitoring plan covering the monitoring period under verification, for approval by the CDM Executive Board)	VVM	201	N/A	OK	OK
c Are there any monitoring aspects of the project activity that are not specified in the methodology, particularly in the case of small-scale	VVM	202	Project is registered under VCS voluntary emission market scheme and in the registered PD v.07, under environmental impact (section 5) PP has	CL14	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
methodologies (e.g. additional monitoring parameters, monitoring frequency and calibration frequency)?			declared compliance with relevant national laws and in line with this committed to: - Release at least 330 lt/sec of water regularly; - Proper handling of excavation wastes; - Proper handling of domestic, solid and liquid wastes; Please provide evidences to support compliance with these commitments.		
4 Compliance of monitoring with the monitoring plan					
a Have the monitoring plan and the applied methodology been properly implemented and followed by the project participants?	VVM	205	Will be answered after related questions are answered and complete monitoring data is reviewed		
b Have the previous monitoring reports been reviewed?	VVM	205	N/A, since this is the initial verification.	OK	OK
c Where applicable, has the impact of revision in the monitoring plan on the current verification been reviewed ?	VVM	205	N/A	OK	OK
d Does the registered/approved monitoring plan have any description of an illustration to calculate net electricity supplied to the grid by the project activity ?	VVM	205	Section 3.2 contains the formula that is used to calculate the net generation.	OK	OK
e If yes to (d) above, has the verification team verified /confirmed the validity of such illustration with supporting documents ?	VVM	205	Yes.	OK	OK
f Have all parameters stated in the monitoring plan, the applied methodology and relevant CDM Executive Board decisions been sufficiently monitored and updated as applicable, including:	VVM	205			

CHECKLIST QUESTION		Ref.	§	COMMENTS	Draft Concl	Final Concl
i	Project emission parameters?	VVM	205	PD has a value of 8692 W/m2 and therefore in line with the guidance of the applied methodology project emissions are neglected.	OK	OK
ii	Baseline emission parameters?	VVM	205	Baseline parameters have been calculated ex-ante in the registered PD and are valid for the duration of the crediting period.	OK	OK
iii	Leakage parameters?	VVM	205	Leakage is neglected in line with the guidance of the applied methodology	OK	OK
iv	Validation of entire procedure of apportioning, if applicable	VVM	205	N/A	OK	OK
v	Management and operational system: the responsibilities and authorities for monitoring and reporting are in accordance with the responsibilities and authorities stated in the monitoring plan?	VVM	205	Implementation is in line with the monitoring plan as verified through the site visit	OK	OK
g	Is the accuracy of equipment used for monitoring in accordance with the relevant guidance provided by the CDM Executive Board and are equipment controlled and calibrated in accordance with the monitoring plan?	VVM	205	Equipment are appropriate and their control and maintenance as well as calibration responsibilities belong solely to TEIAS.	OK	OK
i	Are monitoring results consistently recorded as per approved frequency?	VVM	205	Will be answered after data submission is completed		OK
ii	Have quality assurance and quality control procedures been applied in accordance with the monitoring plan monitoring plan?	VVM	205	Will be answered after data submission is completed		OK
iii	Has the verification team confirmed whether the applicability and correct implementation of any procedure that replaces direct calibration of meters, and any procedure that leads to calculation of parameters used in the ER	VVM	205	N/A	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
determination ?					
5 Assessment of data and calculation of greenhouse gas emission reductions					
a Is a complete set of data for the specified monitoring period is available? (If no, i.e., only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, the DOE shall opt to either make the most conservative assumption theoretically possible in finalizing the verification report, or raise a request for deviation prior to submitting request for issuance, if appropriate).	VVM	208	Will be answered after data submission is completed		OK
b Has information provided in the monitoring report been cross-checked with other sources such as plant log books, inventories, purchase records, laboratory analysis?	VVM	208	Will be answered after data submission is completed		OK
c Have calculations of baseline emissions, proposed CDM project activity emissions and leakage, as appropriate, been carried out in accordance with the formulae and methods described in the monitoring plan and the applied methodology document?	VVM	208	Will be answered after data submission is completed		OK
d Have any assumptions used in emission calculations been justified?	VVM	208	Will be answered after data submission is completed		OK
e Have appropriate emission factors, IPCC default values and other reference values been correctly applied?	VVM	208	Will be answered after data submission is completed		OK

Table 2 Resolution of Corrective Action / Forward Action / Clarification Requests.

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1/2/3/4	Summary of project owner response	Validation team conclusion
CAR01 Applied methodology is given as ACM0002 v.12.2.0, which is inconsistent with the project PD v07. Please revise.	Table 1: 1.e.a	The project PD v07 includes the methodology number of ACM0002 version 11. ACM0002 v.12.2.0 is written by mistake at the Monitoring Report version 1, and it is changed in Monitoring Report version2. Response 2: Reference link for ACM0002 (footnote 1) and the methodology number under Section 4.2 have been corrected accordingly.	<u>Review1:</u> v.12.2.0 still referred to under Section 4.2 and in footnote 1, please revise. <u>CAR remains open</u> <u>Review2:</u> Erroneous references to v.12.2 were corrected. <u>CAR is closed</u>
CAR02 Section includes procedures, etc. only for electricity generated parameter. Please also include procedures, roles and responsibilities for the other parameters that are monitored	Table 1: 1.m.b	Other parameters to be monitored are also included in Section 3.3 of the revised MR. Response 2: 1)- The following explanation has been added under Section 4.2: “(Reservoir area at 1638 m altitude. Please see the technical drawing	<u>Review1:</u> 1. <u>Please indicate the mentioned possible maximum water level value with its source, at which the Api value used, is realized.</u> 2. <u>Please indicate a source for the technical drawing utilized.</u> 3. <u>Monitoring procedures indicated in Table 1 are different from the method applied. Please revise.</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1/2/3/4	Summary of project owner response	Validation team conclusion
		provided under Appendix 1) The drawing is obtained from the project plan prepared by EN-SU Muhendislik which is the company's consultant for etude and design of the project. 2)- The technical drawing which shows the regulator area is obtained from the Final Design documents prepared by EN-SU Muhendislik. 3)- Table 1 has been corrected in line with the Monitoring Plan. 4)- The monitoring procedure for APJ has been deviated. Therefore, the following explanation has been added under Section 2.2. 5)- Corrected accordingly. 6)- Corrected accordingly. <u>Response 3:</u> 2)- Please find attached drawings regarding the calculation of regulator area. The regulator area is calculated based on the technical drawing	4. <u>Monitoring procedure for Apj has deviated from the one indicated in the registered PD, therefore please indicate this deviation under section 2.2 of MR with its justification.</u> 5. <u>Page 12 mentions signed protocols, which is not applicable. Please report meter reading procedures as they are applied;</u> 6. <u>Description of the monitoring plan (section 3.3 of MR v.02) mentions meter calibration responsibilities for personnel. This is contradictory with calibration procedures indicated, please clarify</u> <u>CAR remains open</u> <u>Review2:</u> 1. <u>Design altitude for maximum water level has been indicated and the source was identified. OK.</u> 2. <u>Please indicate the specific source of the referred technical drawing and justify if the source is from approved project documentation.</u> 3. <u>Table has been revised to reflect the adopted practices.</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1/2/3/4	Summary of project owner response	Validation team conclusion
		prepared by EN-SU Muhendislik at the final design phase.	4. <u>Deviation has been reported under 2.2 in MR v.3</u> 5. <u>Description of monitoring revised to reflect the implementation.</u> 6. <u>Redundant reference to meter calibration responsibilities were removed.OK.</u> <u>CAR remains open</u> <u>Review3:</u> 2. <u>Source has been identified as third party engineering company ENSU, OK.</u> <u>CAR is closed.</u>
CAR03 Kindly clarify and ensure consistency of methodology for monitoring EG_{Facility, y} parameter, since various references in the MR v.01, define Monthly Meter Reading Protocols and PMUM records as main source of data and cross-check source inconsistently. Please mind that consistency	Table 1: 1.n.a	EG _{Facility,y} is consistent of the methodology in MR v.02. And also, the extra explanation about working principles of PMUM is added in MR v.02.	<u>Review1:</u> <u>Main source of EG data is indicated as meter reading protocols to be crosschecked with PMUM records consistently</u> <u>CAR is closed.</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1/2/3/4	Summary of project owner response	Validation team conclusion
with the registered project PD v.07 must also be minded.			
CAR04 Please provide Monthly Meter Reading Protocols for verification.	Table 1: 1.o.a	Since the generated electricity is measured remotely, not Monthly Meter Readings are filled in and signed. In place of these records, the measured electricity is provided to the company by email with an excel sheet called "OSF Form". Therefore, the ERs are calculated based on PMUM Records and cross-checked with OSF forms. Response 2: 1)- Remote reading of meters started in May 2011. There is no record for the start date of remote reading. 2)- Because of the motion for stay of execution, the company did not generate electricity during July 2011. After the positive result of the process, the company restarted to generate electricity in August 2011. 3)- Please find attached	<u>Review1:</u> <u>Please clarify when remote reading has started with objective evidence;</u> <u>Please clarify why no generation has occurred on July 2010;</u> <u>Kindly provide the meter reading record for February 2011;</u> <u>Kindly provide the meter reading record for April 2011;</u> <u>Kindly provide the meter reading record for August 2011;</u> <u>CAR remains open</u> <u>Review2:</u> <u>OSF forms which are related to remote reading were provided for 2010 too, before the indicated remote reading start date. Please clarify;</u> <u>OK.</u> <u>Provided, OK.</u> <u>Provided, OK.</u> <u>Provided, OK.</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1/2/3/4	Summary of project owner response	Validation team conclusion
		4)- Please find attached 5)- Please find attached <u>Response 3:</u> There are not any signed documents for the measured electricity before and after the remote reading start date. TEIAS made the readings at the project site before the start of the remote reading and provided the OSF forms to the plant engineer. After the start of the remote reading, the readings were made remotely and OSF forms were sent to the plant engineer by email.	<u>CAR remains open</u> <u>Review3:</u> 1. <u>Justification accepted as same form of transition has been seen in many projects.</u> <u>CAR is closed.</u>
CAR05 Please provide an excel worksheet clearly depicting electricity generation values with the relevant source, any formulae applied etc.	Table 1: 1.o.a	Please see the attached excel sheet regarding the electricity generation and emission reduction <u>Response 2:</u> 1)- The exact electricity generation figures are obtained from OSF forms (and PMUM records) and recorded to the ER calculation spreadsheet. The	<u>Review1:</u> <u>Excel worksheet has been provided, however:</u> 1. <u>Please refrain from rounding values and use number of decimal points identical to the sources used;</u> 2. <u>Please revise the decimal notations (use of '.'s and ', 's)</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1/2/3/4	Summary of project owner response	Validation team conclusion
		values are not rounded up. It is just about the floating point format of excel. Calculations are made with the exact values. 2)- The consistency of use of points and commas has been checked and revised accordingly.	<u>CAR remains open</u> <u>Review2:</u> <u>Value visibility and notations corrected.</u> <u>CAR is closed.</u>
CAR06 Title of the project in the validated PD ver.07 and the monitoring report cover are inconsistent, please revise.		The title of the project has been revised in line with the validated VCS PD v7. The mechanical and electrical installed capacities of the project have been stated in the MR clearly.	<u>Review:</u> <u>Title has been corrected for conformity with the validated MWm value of the project, rounded down conservatively.OK</u> <u>CAR is closed.</u>
CL01 Please Indicate How The Project Reduces Greenhouse Gasses, In Further Detail.	TABLE 1: 1.A.B.i	The explanation about Greenhouse Gasses reduction is added Monitoring Report v.0.2 under Section 1.1.	<u>Review1:</u> <u>It is included under section 1.1 that the project helps reduce GHGs by replacing GHG producing generation sources with a renewable source.</u> <u>CL request is closed.</u>
CL02	Table 1: 1.a.b.ii	Information on implemented technology and installed hardware is	<u>Review1:</u> <u>Info on the project tech has been</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1/2/3/4	Summary of project owner response	Validation team conclusion
Please provide information on implemented technology and installed hardware.		added on Monitoring Report v.0.2 under Section 1.1.	<u>included in line with the validated project PD.</u> <u>CL request is closed.</u>
CL03 Please briefly indicate dates of major events in the project's implementation.	Table 1: 1.a.b.iii	The major event's dates are written under Section 1.1 at Monitoring Report v.0.2.	<u>Review1:</u> <u>The project start date, start date of the crediting period, licensing date and commissioning dates of both phases have been provided under section 1.1</u> <u>CL request is closed.</u>
CL04 Kindly include information on the amount of emission reductions achieved during this monitoring period.	Table 1: 1.a.b.iv	Information about the amount of emission reductions achieved during the monitoring period is added in Monitoring Report v.0.2.	<u>Review1:</u> <u>Amount of emission reductions given in section 1.1 and are baseline emissions inconsistent, please clarify</u> <u>CL request is closed.</u>
CL05 As indicated by the reporter, GAIA is not a project proponent. Kindly move GAIA under section 1.4 of	Table 1: 1.b.a	GAIA is removed from the section 1.4 of the monitoring report v.02.	<u>Review1:</u> <u>Section 1.4 is missing from the monitoring report template, and section numbers are not in line with MR template v.3.1, please review the section numbers for compliance with</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1/2/3/4	Summary of project owner response	Validation team conclusion
the monitoring report v.01			<u>the template.</u> <u>CL request is closed.</u>
CL06 Please provide information on any major overhaul activities and kindly indicate any critical equipment exchange that was performed during this monitoring period.	Table 1: 1.i.b.ii	There have been no major overhaul activities or critical equipment exchange during the monitoring period. This information is added under Section 2.1 at Monitoring Report v.0.2. <u>Response 2:</u> Meter readings have been made remotely during the Monitoring period. The following explanation has been added to the revised MR: <i>“An excel sheet called “OSF Form – Dengeleme Birimi Sayaç Değerleri Formu” which includes the hourly electricity generations and imports is sent to the plant engineer from TEIAS after the remote measurement. If there is a network problem and it is not possible to measure the electricity remotely, a representative from TEIAS visits the</i>	<u>Review1:</u> <u>Please clarify how the meter readings are performed (on-site or remotely). And if they are remotely performed please provide information on the OSOS implementation.</u> <u>CL request remains open</u> <u>Review2:</u> <u>Information on OSOS implementation has been provided.</u> <u>CL request is closed.</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1/2/3/4	Summary of project owner response	Validation team conclusion
		<i>project site and measures the electricity by connecting to the meters directly. However, since there was not a network problem during the Monitoring period, all measurements were made remotely"</i>	
CL07 Kindly revise parameter EG_{Facility, y} to include information on calibration procedures for the monitoring devices.	Table 1: 1.n.c.ii	The TEIAS is responsible of calibration for the electricity meters and the meters were sealed by TEIAS. Extra information about the calibration procedures of the monitoring devices are added under Section 3.2 at Monitoring Report v.0.2. <u>Response 2:</u> The information provided under Section 3.3 has been revised. The electrical engineers are not responsible for calibration.	<u>Review1:</u> <u>Information has been added under section 3.2, however section 3.3 indicates calibration responsibilities for electrical engineers on multiple instances. Please clarify what these calibration responsibilities are for.</u> <u>CL request remains open.</u> <u>Review2:</u> <u>Redundant responsibility removed</u> <u>CL request is closed.</u>
CL08 Please provide a source for the data provided for the A_{PJ} parameter's monitored value to verify its	Table 1: 1.n.c.iii	A_{PJ} parameter is monitored based on technical drawing of the regulator area. The technical drawing of the regulator area obtained from final design of the project is computerized	<u>Review1:</u> <u>Please indicate the project flood discharge level corresponding to the area justification.</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1/2/3/4	Summary of project owner response	Validation team conclusion
validity. Kindly justify the use of 1638 elevation.		(opened with Autocad) and the potential lake area for the flood discharge (possible maximum water level). The lake area is calculated by this computer program as 7,275.54 m². <u>Response 2:</u> The project flood discharge level has been indicated in the revised Monitoring Report. <u>Response 3:</u> Please see FSR report dated March 2009, p.1-2. The floodgate elevation for Incesu weir is 1638 m. Also, the technical drawing at page 394/400 of the pdf file(FSR) can be checked to verify the flood discharge level (floodgate elevation).	<u>CL request remains open</u> <u>Review2:</u> <u>Please support the provided specification with approved project documentation.</u> <u>CL request remains open.</u> <u>Review3:</u> <u>Maximum operation water level for the project has been verified as 1638 through approved project documentation, also supported by the provisional acceptance protocol. Therefore reservoir surface area at this height is conservatively the maximum area, OK.</u> <u>CL request is closed.</u>
CL09 Kindly report value of monitored parameter for Cap_{PJ}	Table 1: 1.n.c.i	The monitored values for Cap_{PJ} and A_{PJ} are provide under Section 3.2 and 3.3 of the revised MR.	<u>Review1:</u> <u>CapPJ has been reported as 62.152, in line with the value validated through document</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1/2/3/4	Summary of project owner response	Validation team conclusion
and A_{PJ} in line with the provided frequency			<u>review and site visit; A_{PJ} has been conservatively assumed as the maximum of the weir design elevation.</u> CL request is clo-sed.
CL10 Reference to ACM0002 in page 9 of MR v01 contains an version value that is inconsistent with the applied version of the methodology, please clarify. Please ensure that this inconsistency is revised throughout the Monitoring Report	Table 1: 1.m.b	The version number of ACM0002 in the page of 9 is changed and it is consistent with the other places that refer methodology in Monitoring Report v.02. <u>Response 2:</u> Please see repose to CAR 01.	Review1: Please see CAR01 CL request remains open. Review2: Closed as CAR01 is closed.
CL11 Parameter defined is $EG_{Facility, y}$, y being the year of the value. Therefore please provide calculations and results for each year in the monitoring period	Table 1: 1.o.a	The calculations and results about the Baseline Emissions were done separately in Monitoring Report v.0.2 under Section 4.1.	<u>Review1:</u> <u>Baseline emissions and net electricity supplied values have been provided annually</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1/2/3/4	Summary of project owner response	Validation team conclusion
separately for verification consideration.			<u>CL request is closed.</u>
CL12 A table was provided in section 4.1 of MR v01. However, table title and the footnote of page 14 have contradicting information with regards to the source of the data. Please clarify	Table 1: 1.o.b	The footnote is deleted in Monitoring Report v.02. There is no contradiction between table title and footnote at version 02. The correct statement is 'PMUM Records of Uzundere I'	<u>Review1:</u> <u>Contradicting information has been removed.</u> <u>CL request is closed.</u>
CL13 Please review sentence " <i>Since baseline emission calculations were realized in the ex-ante manner...</i> "	Table 1: 1.o.a	The sentence of ' <i>Since baseline emission calculations were realized in the ex-ante manner...</i> ' is revised and deleted in Monitoring Report v.02.	<u>Review1:</u> <u>OK.</u> <u>CL request is closed.</u>
CL14 Project is registered under VCS voluntary emission market scheme and in the registered PD v.07, under environmental impact (section 5) PP has declared compliance with relevant national laws and in line with this committed to: ▪ Release at least 330 lt/sec of water regularly; ▪ <u>Proper handling of excavation wastes;</u>	Table 1: 3.c	- The monthly water values are measured and declared by DSI. The Excel Sheet about these average values of water is at the attachment. - Please see attached documents regarding the disposal of wastes. <u>Response 2:</u> There is no available flow data for	<u>Review1:</u> <u>Provided data starts from the date 01/10/2011. Please provide the flow data for the monitoring period being reported.</u> <u>Please indicate the source of the provided map for excavation wastes site.</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1/2/3/4	Summary of project owner response	Validation team conclusion
Proper handling of domestic, solid and liquid wastes; Please provide evidences to support compliance with these commitments.		before 01/10/2011, since the flow data has been recorded starting from this date upon request of DSI. There was not any River Observation Station at the regulator area before this date. The provided map for excavation wastes site was obtained from the Final EIA Report of the project. <u>Response 3:</u> Please find attached documents to verify the installation date of the AGI.	<u>Waste oil disposal was performed by licensed companies on 10/06/2010 and 06/04/2012 in accordance with local regulations; a contract for the collection of domestic wastes has been verified and is still valid; and sanitary waste disposal was verified through receipts.</u> <u>CL request remains open</u> <u>Review2:</u> <u>Please provide a justification for the installation date of the AGI station.</u> <u>CL request remains open.</u> <u>Review3:</u> <u>Correspondence between DSI and PP show that the implementation of AGI stations have been completed in phases during the summer of 2011 and water release data are available after 01/10/2011.</u> <u>CL request is closed.</u>
ITR FINDINGS:			

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1/2/3/4	Summary of project owner response	Validation team conclusion
CL15 Project title in the MR is not in line with the validated project PD and validation report. Please revise.		MR has been corrected accordingly.	<u>Review:</u> <u>Project title was revised for consistency with the validated PD and FVR.</u> <u>CL request is closed.</u>
CL16 In MR section 2.1, the date of construction and the continued operation periods are missing, please include.		Please see the statements added to Section 2.1 of the revised MR.	<u>Review:</u> <u>Section 2.1 was revised to indicate the start of construction as 'early 2007'</u> <u>CL request is closed.</u>
CAR07 In the excel work book: 1. Please show all calculations with formulae in the workbook, values have been entered manually. 2. Net electricity supplied values for May 2010 and Oct. 2010 are incorrect. 3. Total emission reductions for both 2010 and 2011 are incorrect, please revise		1)- Hard entered values have been replaced with the related formulas. 2)- Corrected accordingly. 3)- Corrected accordingly.	<u>Review:</u> 1. <u>Excel sheet has been repaired to show calculations.</u> 2. <u>Values were corrected, OK.</u> 3. <u>Revised as the sheet is corrected.</u> <u>CAR is closed.</u>
CAR08		1)- Corrected as "Market financial Settlement Center (PMUM) records" 2)- Corrected as "Crosscheck with on-site	<u>Review:</u> 1. <u>'source of data' corrected,</u> 2. <u>Corrected, OK;</u> 3. <u>corrected, OK</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in table 1/2/3/4	Summary of project owner response	Validation team conclusion
The validated PD indicates that PMUM records are the main source for generation data, which are to be crosschecked with on-site recordings. In the MR: 1. section 3.2 'source of data' and 2. 'QA/QC procedures' for EGfacility,y source states otherwise; 3. Section 3.3 states PMUM as source of crosscheck, not the main source; 4. Please check excel worksheet also for source of generation data		recordings" 3)- Corrected as "on-site measurement records" 4)- The source of generation data used in the excel work sheet is PMUM.	4. <u>indicated as PMUM, OK</u> <u>CAR is closed.</u>
CL17 Please include in the MR, a comparison of the realized emission reductions and estimated emission reductions in the validated PD, for the duration of the monitoring period.		Comparison of the realized and estimated emission reductions has been discussed in Section 4.4 of the revised MR.	<u>Review:</u> <u>A comparison was included where the realized values are below the estimated (about 85% of expected). Relevant justifications were included and found acceptable.</u> <u>CL request is closed.</u>